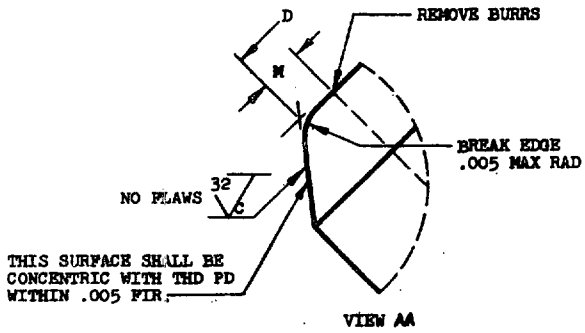
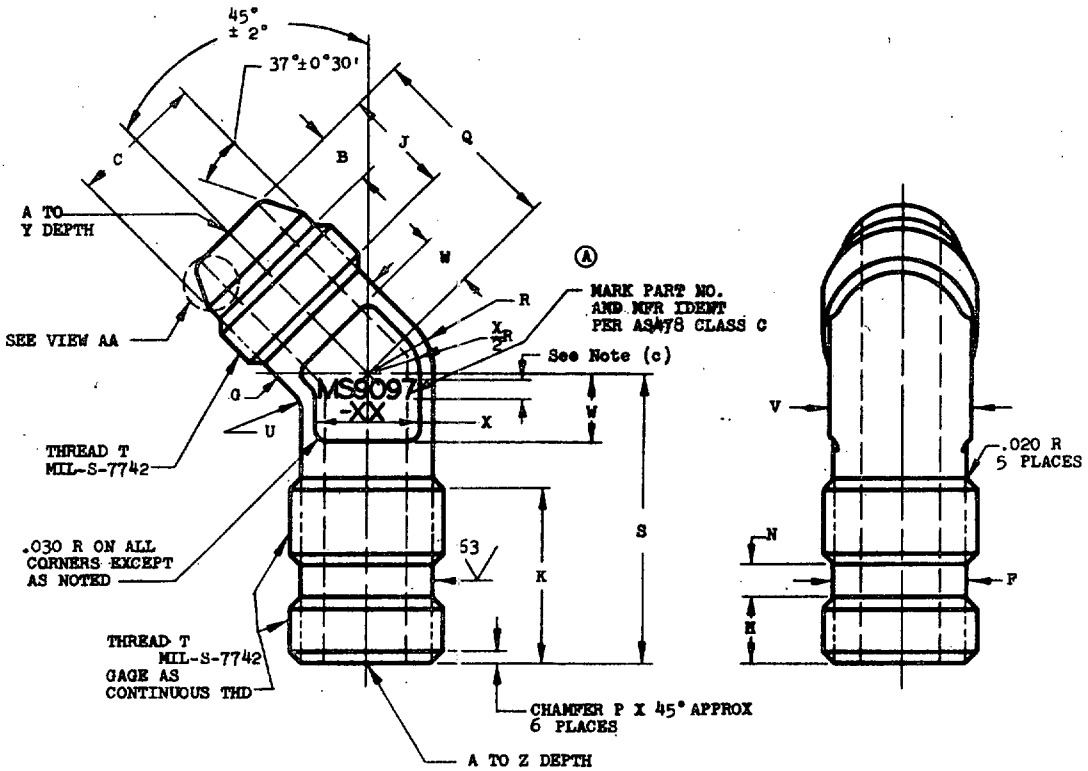
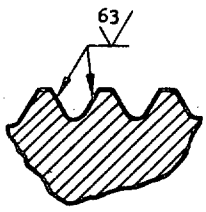
MS9097(ASG)

FED.SUP CLASS.
4730







THIS STANDARD WAS DEVELOPED COOPERATIVELY WITH THE ENGINE AND PROPELLER UTILITY PARTS COMMITTEE OF THE SAE.

APPROVED 11 Apr 60 REVISED Ⓐ 11 May 62

CUSTODIAN USAF - AFSC Navy - BuWeps	MILITARY STANDARD	MS9097(ASG)
PROCUREMENT SPECIFICATION NONE	ELBOW, TUBE - AMS 4135, BOSS, 45°	SHEET 1 OF 2

MS FORM 1 NOV 56 223

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PART NO.	(A)(f) E	NOM TUBE OD	THREAD T	A DIA	B +.015 -.000	C DIA +.000 -.010	D DIA ±.003	F DIA +.002 -.003 (a)(b)	G DIA	H ±.010	J ±.010	K ±.010
MS9097-08	.080	.500	.750-16UNF-3A	.387- .395	.229	.660	.462	.660	.650	.312	.489	.903
MS9097-09	.090	.562	.8125-16UN-3A	.434- .442	.229	.722	.509	.722	.710	.312	.489	.903
MS9097-10	.090	.625	.875-14UNF-3A	.480- .488	.255	.773	.555	.773	.765	.357	.585	1.018
MS9097-11	.100	.688	1.000-12UNF-3A	.542- .552	.304	.882	.619	.882	.870	.417	.663	1.122
MS9097-12	.100	.750	1.0625-12UN-3A	.604- .614	.304	.945	.681	.945	.935	.417	.663	1.122
MS9097-14	.100	.875	1.1875-12UN-3A	.729- .739	.304	1.070	.806	1.070	1.055	.417	.710	1.122
MS9097-16	.120	1.000	1.3125-12UN-3A	.839- .851	.312	1.195	.918	1.195	1.185	.417	.720	1.122
MS9097-18	.140	1.125	1.500-12UNF-3A	.948- .960	.312	1.382	1.027	1.382	1.365	.417	.767	1.122
MS9097-20	.150	1.250	1.625-12UN-3A	1.073-1.086	.363	1.507	1.153	1.507	1.495	.417	.767	1.122
MS9097-24	.160	1.500	1.875-12UN-3A	1.307-1.320	.374	1.756	1.387	1.756	1.745	.417	.889	1.122
MS9097-28	.210	1.750	2.250-12UN-3A	1.542-1.557	.447	2.131	1.652	2.131	2.120	.417	1.017	1.205
MS9097-32	.210	2.000	2.500-12UN-3A	1.776-1.791	.457	2.381	1.886	2.381	2.370	.417	1.142	1.320

PART NO.	M MIN	N +.010 -.000	P	Q ±.010	R RADIUS	S ±.010	U RADIUS	V	W	X	Y ±.010	Z ±.010	APPROX WEIGHT LBS/EA
MS9097-08	.030	.187	.040-.070	.945	.325	1.375	.094	.688	.310	.430	1.023	1.406	.058
MS9097-09	.030	.203	.040-.070	.945	.355	1.375	.094	.750	.310	.440	1.023	1.406	.066
MS9097-10	.030	.219	.050-.080	1.083	.387	1.531	.094	.812	.350	.500	1.177	1.625	.085
MS9097-11	.030	.234	.050-.080	1.255	.435	1.734	.094	.938	.420	.500	1.364	1.859	.126
MS9097-12	.030	.234	.050-.080	1.255	.467	1.734	.094	1.000	.420	.500	1.364	1.859	.137
MS9097-14	.030	.234	.050-.080	1.440	.527	1.734	.125	1.125	.420	.500	1.534	1.859	.166
MS9097-16	.030	.234	.050-.080	1.450	.592	1.734	.125	1.250	.420	.500	1.544	1.859	.195
MS9097-18	.030	.234	.050-.080	1.574	.682	1.750	.125	1.438	.450	.600	1.699	1.921	.276
MS9097-20	.030	.234	.050-.080	1.574	.750	1.750	.125	1.625	.450	.600	1.699	1.921	.308
MS9097-24	.030	.234	.050-.080	1.762	.872	1.750	.125	1.812	.450	.600	1.855	1.921	.395
MS9097-28	.044	.234	.050-.080	1.949	1.062	1.969	.188	2.188	.500	.660	2.199	2.188	.678
MS9097-32	.044	.234	.050-.080	2.199	1.188	2.203	.250	2.438	.500	.750	2.480	2.422	.885

(a) DIAMETER F SHALL BE FREE OF THREAD MARKS.
(b) DIAMETER F SHALL BE CONCENTRIC WITH PD OF THREAD WITHIN .0025 FIR.
(c) .040-.060 FOR -08 THRU -16 SIZES; .090-.150 FOR SIZES -18 AND LARGER SIZES.
(d) ALL DIAMETERS SHALL BE CONCENTRIC WITHIN .010 FIR. UNLESS OTHERWISE SPECIFIED.
(A)(e) PARTING LINE MISMATCH .015 MAX.
(A)(f) E MIN WALL THICKNESS BETWEEN FORGED SURFACE AND HOLE A.

MATERIAL: ALUMINUM AMS 4135
FINISH: ANODIC TREATMENT AMS 2470.
SURFACE ROUGHNESS: AS 291. UNLESS OTHERWISE SPECIFIED MACHINED SURFACES TO BE 125 MICROINCHES.
PARTS SUBJECT TO FLUORESCENT PENETRANT INSPECTION PER AMS 2645.
BREAK SHARP EDGES .003-.015 UNLESS OTHERWISE SPECIFIED.
DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED: TOLERANCES; LINEAR DIMENSIONS ±.015, DIMENSIONS ARE TO SHARP CORNERS UNLESS OTHERWISE SPECIFIED.
DO NOT USE UNASSIGNED PART NUMBERS.
AS, & AMS ARE SAE PUBLICATIONS.

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APPROVED 11 APR 60 REVISED (A) 11 May 62

CUSTODIANS
USAF - AFSC
Navy - Bureps

PROCUREMENT SPECIFICATION
NONE

MILITARY STANDARD

ELBOW, TUBE - AMS 4135, BOSS, 45°

MS9097(ASG)

SHEET 2 OF 2

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